

Requirements Engineering & UML Use-Case Modelling

PES1UG24CS144-Clinical Trial Subject Recruitment Engine

Actors: Trial Candidate, Clinical Researcher, Notification Service, Compliance Auditor.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compare a Trial Candidate's questionnaire answers against the inclusion and exclusion criteria of the selected trial and mark the candidate as Eligible or Not Eligible.	High	Pass: A candidate who meets an exclusion criterion is marked Not Eligible and the failed criteria are listed. Fail: Such a candidate is marked Eligible.	Automated pre-screening is the main purpose of the system and prevents unsafe enrolment.
FR-002	Functional	The system shall allow a Clinical Researcher to add and edit the inclusion and exclusion criteria of a trial.	High	Pass: Criteria saved by the researcher are used in the very next screening. Fail: Screening still uses the old criteria.	Trial criteria are revised during recruitment and must stay up to date.
FR-003	Functional	The system shall display the informed consent form to an eligible candidate and store the candidate's electronic signature along with the form version, date and time.	High	Pass: The consent record contains the signature, form version and timestamp, and enrolment is blocked until it exists. Fail: A candidate is enrolled without a stored consent record.	Informed consent is legally required before a subject can join a trial.
FR-004	Functional	The system shall allow a consented candidate to book study visits and shall issue an automated reminder before each booked visit. The system shall allow the Clinical Researcher to mark each visit as Attended or Missed.	Medium	Pass: A reminder is dispatched for every booked visit, and a visit the subject did not attend appears as Missed in the visit compliance log. Fail: No reminder is dispatched, or the visit is shown as Attended or is not recorded at all.	Visit compliance has to be tracked to detect protocol deviations, and reminders reduce missed visits.
FR-005	Functional	The system shall show the Clinical Researcher and the Compliance Auditor a report giving the number of candidates screened, found eligible, consented and enrolled for a selected trial.	Medium	Pass: Every count in the report matches the stored candidate records. Fail: Any count differs from the stored records.	Researchers need to track recruitment progress against the enrolment target, and auditors need visibility for monitoring.
NFR-001	Non-functional (Security)	The system shall record an audit trail entry containing the user, date, time, old value and new value for every change made to a candidate, eligibility or consent record, as required by 21 CFR Part 11.	High	Pass: Every record change creates an audit entry, and stored entries cannot be edited or deleted. Fail: Any record change leaves no audit entry.	Regulators require proof that clinical records have not been altered.
NFR-002	Non-functional (Performance)	The system shall display the eligibility result within 3 seconds for at least 95% of submissions while 200 candidates are using the system at the same time.	High	Pass: A load test with 200 concurrent users returns 95% of results in 3 seconds or less. Fail: More than 5% of results take longer than 3 seconds.	Slow screening feedback causes candidates to abandon the application.